

REFLECT ON DOMAIN

Matin Mohammadi, Robert Hanssen, Sander Gnanavel

IKT467-1 25H

Obligatorisk gruppeerklæring

Den enkelte student er selv ansvarlig for å sette seg inn i hva som er lovlige hjelpemidler, retningslinjer for bruk av disse og regler om kildebruk. Erklæringen skal bevisstgjøre studentene på deres ansvar og hvilke konsekvenser fusk kan medføre. Manglende erklæring fritar ikke studentene fra sitt ansvar.

1.	Vi erklærer herved at vår besvarelse er vårt eget arbeid, og at vi ikke har brukt andre kilder eller har mottatt annen hjelp enn det som er nevnt i besvarelsen.	Ja / Nei
2.	Vi erklærer videre at denne besvarelsen: • Ikke har vært brukt til annen eksamen ved annen avdeling/universitet/høgskole innenlands eller utenlands. • Ikke refererer til andres arbeid uten at det er oppgitt. • Ikke refererer til eget tidligere arbeid uten at det er oppgitt. • Har alle referansene oppgitt i litteraturlisten. • Ikke er en kopi, duplikat eller avskrift av andres arbeid eller besvarelse.	Ja / Nei
3.	Vi er kjent med at brudd på ovennevnte er å betrakte som fusk og kan medføre annullering av eksamen og utestengelse fra universiteter og høgskoler i Norge, jf. Universitets- og høgskoleloven §§4-7 og 4-8 og Forskrift om eksamen §§ 31.	Ja / Nei
4.	Vi er kjent med at alle innleverte oppgaver kan bli plagiatkontrollert.	Ja / Nei
5.	Vi er kjent med at Universitetet i Agder vil behandle alle saker hvor det forligger mistanke om fusk etter høgskolens retningslinjer for behandling av saker om fusk.	Ja / Nei
6.	Vi har satt oss inn i regler og retningslinjer i bruk av kilder og referanser på biblioteket sine nettsider.	Ja / Nei
7.	Vi har i flertall blitt enige om at innsatsen innad i gruppen er merkbart forskjellig og ønsker dermed å vurderes individuelt. Ordinært vurderes alle deltakere i prosjektet samlet.	Ja / Nei

Publiseringsavtale

Fullmakt til elektronisk publisering av oppgaven Forfatter(ne) har opphavsrett til oppgaven. Det betyr blant annet enerett til å gjøre verket tilgjengelig for allmennheten (Åndsverkloven. §2).

Oppgaver som er unntatt offentlighet eller taushetsbelagt/konfidensiell vil ikke bli publisert.

Vi gir herved Universitetet i Agder en vederlagsfri rett til å gjøre oppgaven tilgjengelig for elektronisk publisering:	Ja / Nei
Er oppgaven båndlagt (konfidensiell)?	Ja / Nei
Er oppgaven unntatt offentlighet?	Ja / Nei

Contents

List of Figures	ii
List of Tables	ii
0.1 Tools	1
1 Concepts	2
1.1 Concepts (UML structure)	2
1.2 How they are related	3
2 Properties	4
3 Snapshots	5
3.1 Snapshots	5
3.1.1 Free-flow mixed traffic	5
3.1.2 AV platoon in dense traffic ($t = 61.0$ s)	6
4 Robustness	8
4.1 Robustness	8
4.2 Changing a Concept	8
5 Initial and final states	9
5.1 Capacity Perspectives	9
5.1.1 Physical Capacity	9
5.1.2 Operational Capacity	10
5.2 Initial and Final States	10
5.2.1 Initial State	10
5.2.2 Final States	11
Bibliography	12

0.1 Tools

- ChatGPT: used for grammar editing and improving writing style
- Overleaf: used for writing and formatting the report

Chapter 1

Concepts

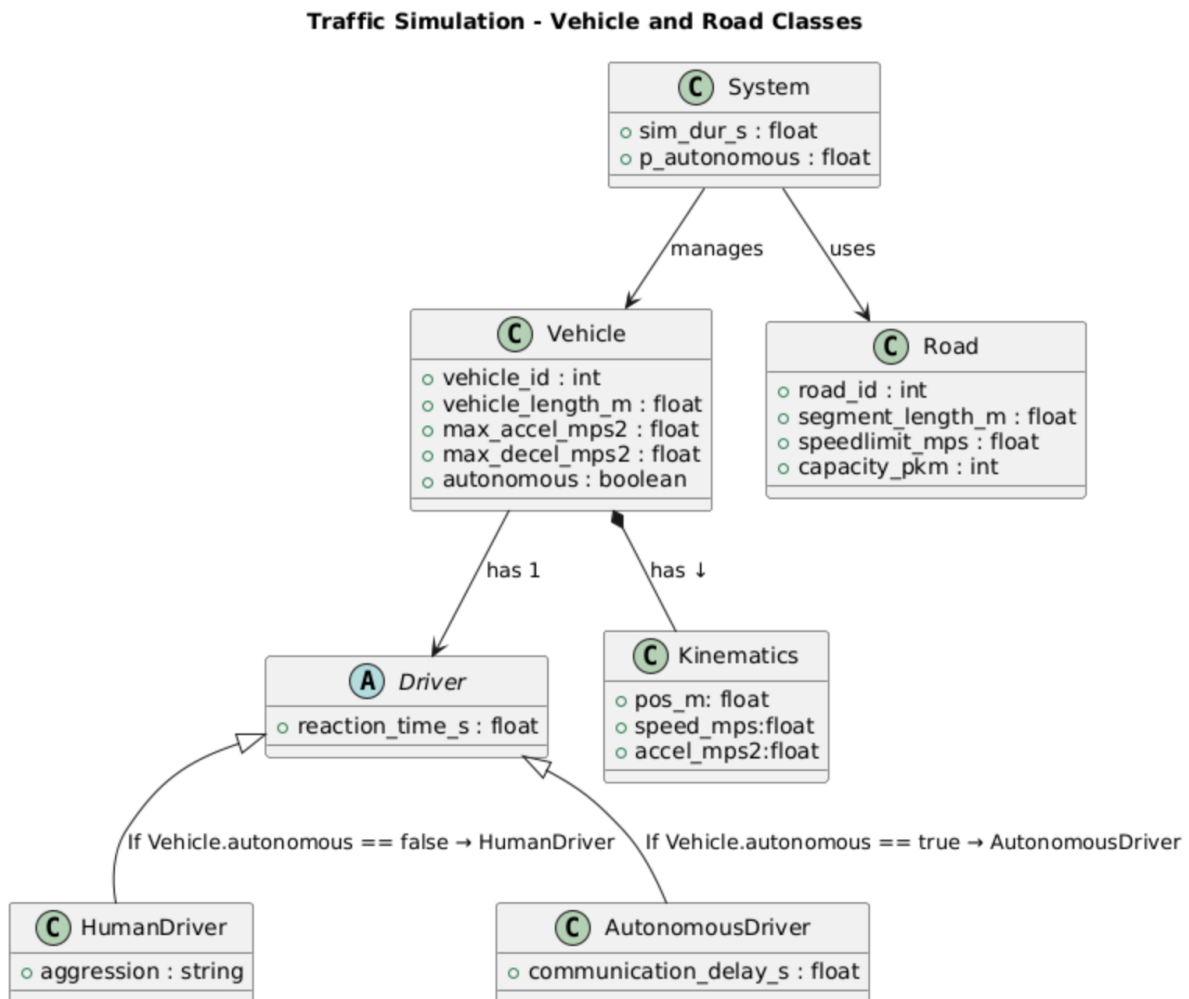


Figure 1.1: UML class diagram of the traffic simulation model `autonomous`.

1.1 Concepts (UML structure)

This model captures a single simulation run **System** operating on a **Road**, populated by multiple **Vehicle** instances. Each vehicle maintains a **Kinematics** state and is controlled by exactly one **Driver** profile, which is either a **HumanDriver** or an **AutonomousDriver**.

System Represents one simulation run with global controls and scenario mix. *Input properties, not stored as System attributes:* `sim_dur_s : float`, `p_autonomous : float`.

Road Defines the traffic environment for the run. *Properties:* `road_id: int, segment_length_m: float, speedlimit_mps: float, capacity_pkm: int`.

Vehicle Models an individual agent traversing the road network. *Properties:* `vehicle_id: int, vehicle_length_m: float, max_accel_mps2: float, max_decel_mps2: float, autonomous: boolean`.

Kinematics Encodes the instantaneous motion state of a vehicle. *Properties:* `pos_m: float, speed_mps: float, accel_mps2: float`.

Driver Base type for driver behaviour and control latency. *Properties:* `reaction_time_s: float`.

HumanDriver (subclass of Driver) Human driving profile capturing behavioural variation. *Properties:* `aggression: string`.

AutonomousDriver (subclass of Driver) Automated controller profile with communication delay. *Properties:* `communication_delay_s: float`.

1.2 How they are related

The system selects the Road configuration and manages the fleet of vehicles. Each Vehicle owns exactly one Kinematics object and is paired with exactly one Driver profile via a simple selection rule.

System → Road (uses) Each simulation run references the road configuration it operates on.

System → Vehicle (manages) The run creates and manages the participating vehicles.

Vehicle — Kinematics (composition) A vehicle owns a single kinematic state object whose lifecycle is bound to the vehicle.

Inheritance **HumanDriver** and **AutonomousDriver** specialize the abstract **Driver** type, sharing `reaction_time_s` while introducing type-specific parameters.

Vehicle → Driver (association; exactly one) Every vehicle has one driver profile.

Constraint: If `Vehicle.autonomous == true` ⇒ the driver is **AutonomousDriver**;
if `Vehicle.autonomous == false` ⇒ the driver is **HumanDriver**.

Chapter 2

Properties

The System controls a single run with `sim_dur_s` as a float in seconds and `p_autonomous` as a float between 0 and 1. We store both to two decimal places. The Road defines the environment with `road_id` as an integer, `segment_length_m` as a float in meters, `speedlimit_mps` as a float in meters per second, and `capacity_pkm` as an integer for vehicles per kilometer. Road length uses one meter precision and speed limit uses one tenth meter per second precision. Each Vehicle has `vehicle_id` as an integer, `vehicle_length_m` as a float, `max_accel_mps2` as a float, `max_decel_mps2` as a float, and `autonomous` as a boolean. Vehicle length uses one tenth meter precision. Acceleration and deceleration use one tenth meter per second squared precision. The Kinematics block holds changing motion for a vehicle with `pos_m` as a float in meters, `speed_mps` as a float in meters per second, and `accel_mps2` as a float in meters per second squared. We record position to one tenth meter, speed to one hundredth meter per second, and acceleration to one hundredth meter per second squared. Driver is an abstract profile with `reaction_time_s` as a float in seconds stored to one hundredth second. `HumanDriver` adds aggression as a string label such as calm or aggressive. `AutonomousDriver` adds `communication_delay_s` as a float in seconds stored to one hundredth second. IDs and booleans are exact by definition. Numeric inputs are accurate to the configured values, and the chosen decimal precision is enough to compute throughput, travel time, delay, and density reliably without producing oversized files.

Chapter 3

Snapshots

3.1 Snapshots

A snapshot represents a specific moment of the model captured during the simulation. Therefore, it is important to focus only on the objects that are actively involved at that time. In these snapshots, we do not include the Driver class directly, since its subclasses, HumanDriver and AutonomousDriver, already inherit all relevant attributes and behaviors from the parent class. The focus is instead placed on the main objects and their interactions

3.1.1 Free-flow mixed traffic

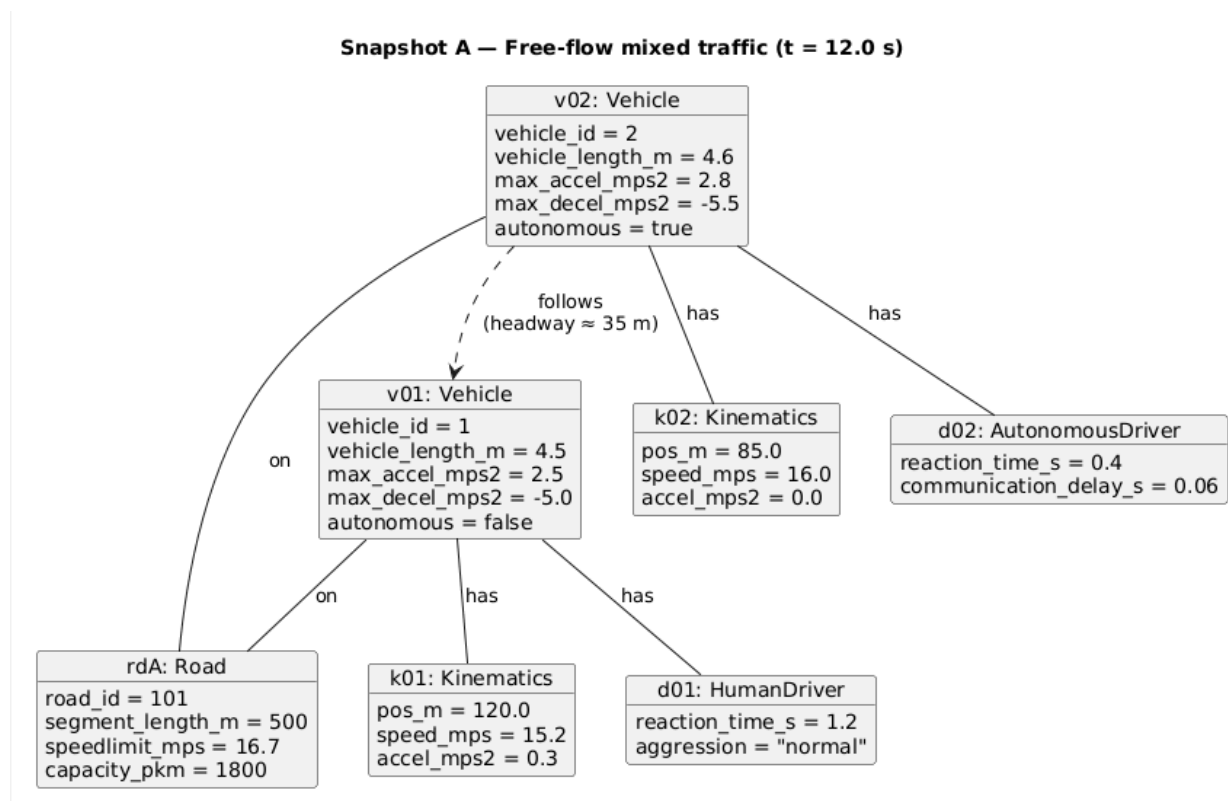


Figure 3.1: Snapshot-1

At time $t = 12.0$ seconds, the simulation shows a stable free-flow traffic state managed by the main system instance (sys1:System). The system controls a single active road segment (rdA:Road) that has a speed limit of 16.7 m/s and a total capacity of 1800 vehicles per km. On this road, two vehicles are present simultaneously: one driven by a human and the other

operating autonomously. The human-driven vehicle (v01:Vehicle) is located at position 120 m with a speed of 15.2 m/s and a slight acceleration of 0.3 m/s², indicating smooth motion just below the road's speed limit. Its corresponding driver object (d01:HumanDriver) has a reaction time of 1.2 seconds and a normal aggression level, representing an average human behavior model. The autonomous vehicle (v02:Vehicle) is positioned 85 m from the start of the same road segment, moving at 16.0 m/s with zero acceleration, suggesting steady cruising behavior. Its controller (d02:AutonomousDriver) is configured with a short reaction time of 0.4 seconds and a communication delay of 0.06 seconds, reflecting faster and more consistent response than a human driver. Both vehicles are traveling safely within the legal speed limit and maintain a headway distance of approximately 35 meters. No braking, congestion, or interaction conflicts occur at this moment, meaning the traffic flow is uniform and uninterrupted. This snapshot represents a reference state of mixed-traffic equilibrium, which later serves as a baseline for analyzing dynamic events such as braking reactions, lane changes, and platoon formation in subsequent simulation stages.

3.1.2 AV platoon in dense traffic ($t = 61.0$ s)

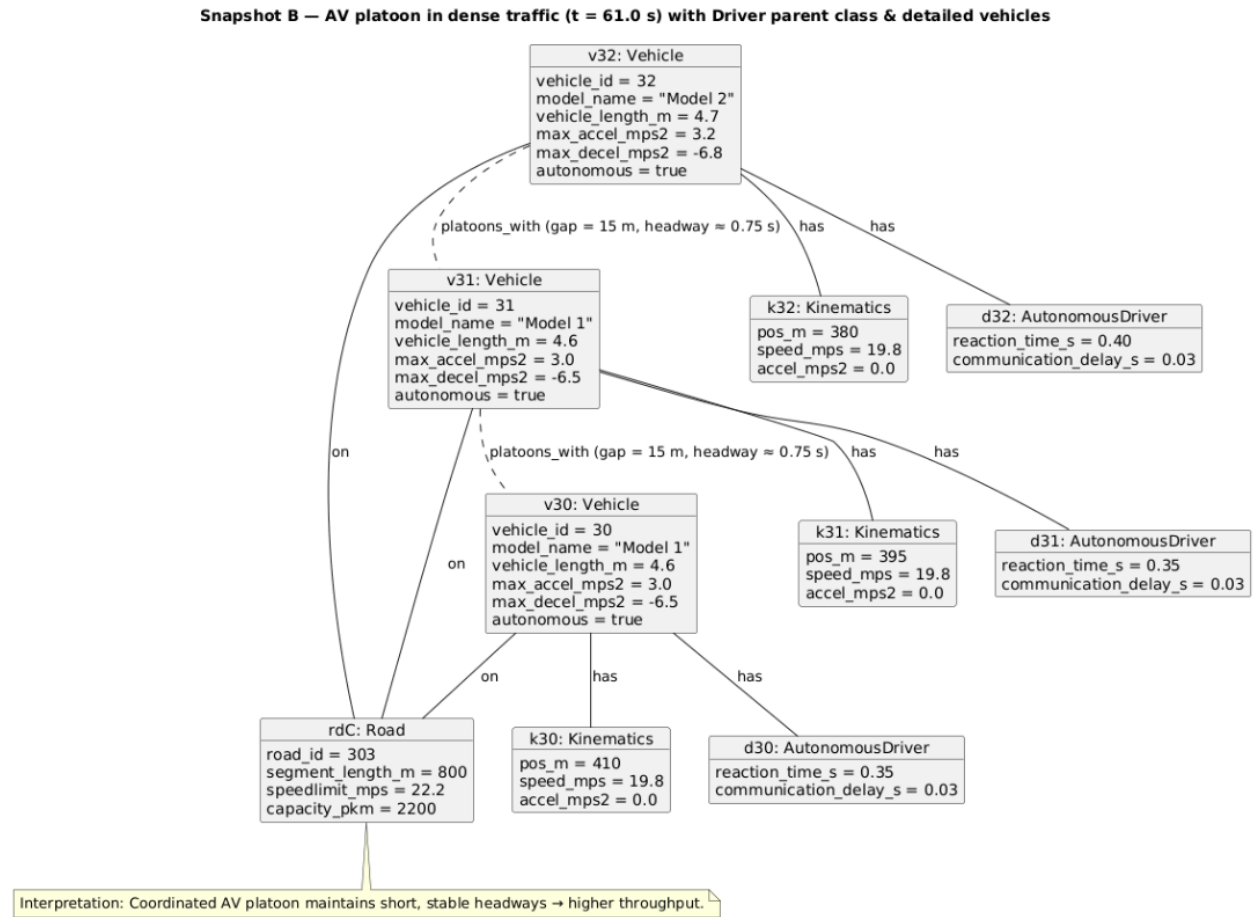


Figure 3.2: Snapshot 2

At $t = 61.0$ seconds, the simulation represents a dense-traffic condition composed exclusively of autonomous vehicles forming a coordinated platoon. The active road segment (rdC:Road) is 800 m long with a speed limit of 22.2 m/s (80 km/h) and an effective capacity of 2,200 vehicles per kilometre, reflecting the improved throughput achievable through close coordination and reduced following distances. The main system instance (sys1:System) supervises three autonomous vehicles (v30, v31, and v32) that travel at uniform speeds and maintain short,

stable headways.

The lead vehicle (`v30:Vehicle`) is positioned at 410 m with a constant speed of 19.8 m/s (71 km/h) and zero acceleration, establishing the reference pace for the platoon. The second vehicle (`v31:Vehicle`) follows at 395 m with the same speed and no acceleration, while the third vehicle (`v32:Vehicle`) maintains identical kinematic values at 380 m. Each vehicle's controller (`d30-d32:AutonomousDriver`) has a reaction time of approximately 0.35–0.40 seconds and a minimal communication delay of about 0.03 seconds, allowing for near-instantaneous synchronization of braking and acceleration across the group.

The spatial gap between consecutive vehicles is approximately 15 m, corresponding to a time headway of about 0.75 seconds at the current speed. Despite this short distance, the platoon remains perfectly stable due to continuous sensor fusion and inter-vehicle communication. No oscillation or stop-and-go behaviour is detected, and the collective movement behaves almost as a single dynamic unit.

This snapshot illustrates how fully autonomous coordination can drastically increase effective lane utilization by reducing unnecessary gaps and ensuring uniform velocity among vehicles. Compared with the previous mixed-traffic scenarios, the autonomous platoon demonstrates smoother flow, shorter response times, and significantly higher traffic throughput without compromising safety. It represents an optimal state of traffic efficiency where automation and communication jointly enable dense yet stable mobility.

Chapter 4

Robustness

4.1 Robustness

In "Understanding Modelling and Programming," a system is defined as a dynamic "set of active objects that interact with each other and with objects in the environment of the system" [1]. Therefore, we evaluate the **robustness** of our traffic simulation model by considering how modifications to its core concepts, as defined in our UML Class Diagram in Figure 1.1, would propagate through the system.

4.2 Changing a Concept

In our UML class diagram the **Vehicle** class has associated with it **Kinematics** and a **Driver** with specified either **HumanDriver** or **AutonomousDriver**. This design is robust with respect to changes in driver types. For instance, if we wanted to introduce a new type of driver, such as a **TruckDriver** with distinct properties we could add a new class that inherits from the abstract **Driver** class. This allows the model to evolve without requiring a fundamental redesign of its central components. Conversely, the model is less robust against changes to its fundamental physical assumptions, such as the dimensionality of movement. If we wanted to include lane-changing behavior, this would represent a shift from a 1D to a 2D perspective. It would require replacing the scalar properties in **Kinematics** with vectors (e.g., for position, velocity, and acceleration) and would be difficult to implement. Additionally, the current model assumes an uninterrupted highway segment. If we wanted to introduce new environmental features like traffic lights or intersections, the model in its current form would not be robust. This would require a fundamental change in the **Driver's** behavior, as drivers would now need to react to external signals, not just to the vehicle in front of them. The **Road** class would also need to be modified to accommodate these new objects.

Chapter 5

Initial and final states

5.1 Capacity Perspectives

Before specifying concrete state snapshots, structural constraints that determine which configurations are physically possible should be defined. The capacity of a road segment is not a single value but depends on the assumed spacing between vehicles. We can define a general formula for capacity based on a generic spacing parameter, d .

The maximum number of vehicles is given by:

$$\text{capacity_pkm} = \frac{L_{\text{road}}}{L_{\text{vehicle}} + d} \quad (5.1)$$

where:

- L_{road} = Length of the road segment
- L_{vehicle} = Average vehicle length
- d = The assumed spacing between vehicles

This capacity can be expressed as a maximum density, ρ_{max} .

$$\rho_{\text{max}} = \frac{\text{capacity_pkm}}{L_{\text{road}}/1000} \quad (\text{veh/km}) \quad (5.2)$$

We can now explore two distinct perspectives by substituting different values for the spacing parameter d .

5.1.1 Physical Capacity

The maximum capacity is calculated from the road length, vehicle lengths, and the minimum distance constraint. This perspective considers the absolute physical limit of the road, using the minimum possible spacing to avoid collision, $d = d_{\text{min}}$.

For a 5 km road assuming a minimal bumper-to-bumper spacing of $d_{\text{min}} = 1$ m:

$$\text{capacity_pkm} = \left\lfloor \frac{5000}{5 + 1} \right\rfloor = \left\lfloor \frac{5000}{6} \right\rfloor = 833.\bar{3} = \underline{\underline{833}} \text{ vehicles} \quad (5.3)$$

with $L_{\text{road}} = 5000$ m, $L_{\text{vehicle}} = 5$ m, and $d_{\text{min}} = 1$ m.

This capacity represents maximum number of vehicles that can physically occupy the road

segment while maintaining minimum spacing, and serves as an upper bound. In practice, traffic flow typically breaks down well before reaching this theoretical maximum due to instabilities in human driving behavior.

Alternatively, we can express capacity as **density**:

$$\rho_{\max} = \frac{\text{capacity_pkm}}{L_{\text{road}}/1000} = \frac{833}{5} = 166 \text{ veh/km} \quad (5.4)$$

5.1.2 Operational Capacity

The point at which the road reaches saturation gives the estimated maximum density of vehicles while maintaining a state of free-flowing traffic. This perspective considers the practical limit for maintaining safe, free-flowing traffic. Here, we use a safe following distance that varies with velocity, $d = d_{\text{safe}}$.

For our earlier example with $L_{\text{road}} = 5000$ m, $L_{\text{vehicle}} = 5$ m, and $d_{\text{safe}} = 50$ m:

$$\text{capacity_pkm} = \left\lfloor \frac{5000}{5 + 50} \right\rfloor = \left\lfloor \frac{5000}{55} \right\rfloor = 90.\overline{909} = \underline{\underline{90}} \text{ vehicles} \quad (5.5)$$

This capacity represents theoretical maximum number of vehicles that can occupy the road segment while maintaining maximum efficiency and traffic-flow. Going beyond this threshold we may see over-saturation leading to a break down of traffic flow and congestion occurring.

This capacity can be expressed as a **density** of:

$$\rho_{\max} = \frac{\text{capacity_pkm}}{L_{\text{road}}/1000} = \frac{90}{5} = 18 \text{ veh/km} \quad (5.6)$$

5.2 Initial and Final States

An execution of a system is a collection of snapshots over time. The initial and final states are the specific snapshots that mark the beginning and end of a simulation run. In our system, each snapshot captures the structural state, kinematic state, and road state at a specific moment. An execution then becomes a sequence of these snapshots, allowing us to observe how the traffic system evolves.

5.2.1 Initial State

The simulation scenarios begin from an ideal state where the system has not yet experienced any disturbances. The characteristics of this initial state is described by the Spatial, Kinematic, and Agent configuration detailed below.

Spatial Configuration:

- Vehicles are distributed uniformly along the road segment.
- Spacing between vehicles is uniform and $>$ minimum distance.

Kinematic Configuration:

- Vehicles have a velocity (`speed_mps`) $\approx \pm 5\%$ of the roads defined.
- Vehicles have an acceleration (`acceleration_mps2`) $= 0$ (steady-state).

Agent Configuration:

- The proportion of HumanDriver vs. AutonomousDriver is set according to the experiment parameters.
- Each driver's behavioral parameters are initialized.

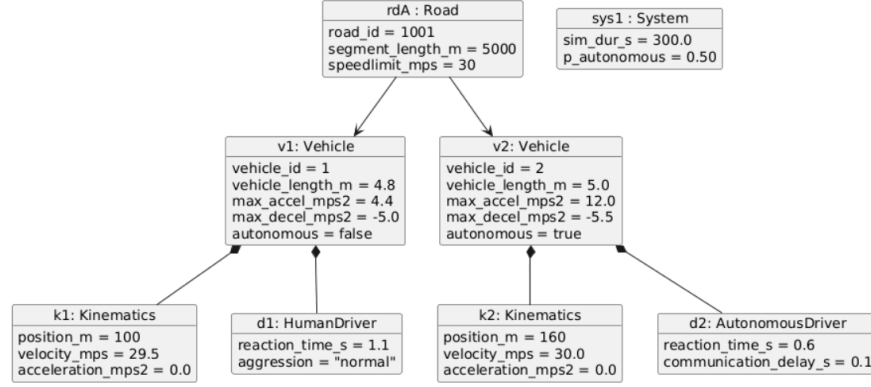


Figure 5.1: UML Object Diagram - Initial State, Free-Flowing Traffic.

5.2.2 Final States

There are several potential final states, depending on the scenario and parameters. In our simulation we do not have a predefined goal state. The "final state" is simply the snapshot of the system at the moment the simulation ends, when `sim_dur_s` is reached.

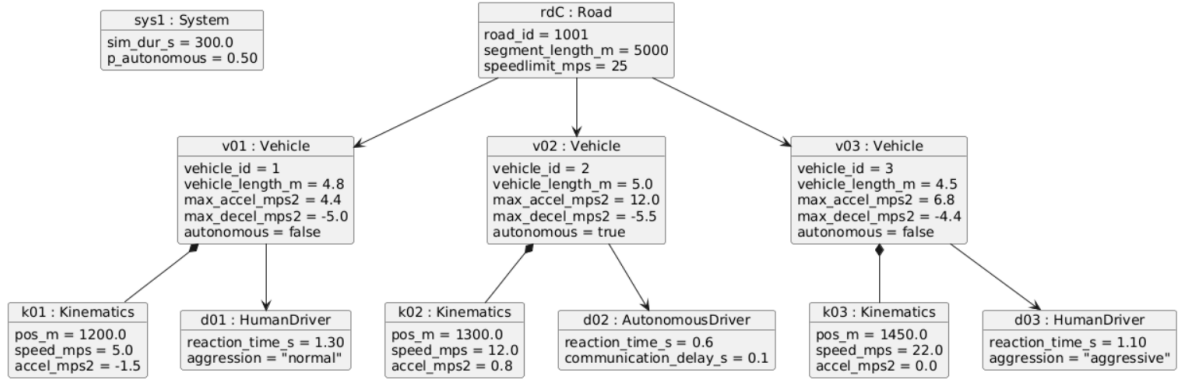


Figure 5.2: UML Object Diagram - Congested Final State ($t = t_{max}$).

In a scenario where congestion has occurred, the final state will show a developed traffic jam. This snapshot would be characterized by a cluster of vehicles having `speed_mps` at or near zero, while vehicles further upstream might still be approaching at high speed. The `capacity_pkm` threshold of the road will have been exceeded in certain sections. In a transient state, the simulation might end while a traffic wave is still propagating. In this case, the final snapshot would capture a system with high variance in vehicle speeds and densities along the road segment.

The optimal "final state" will bear resemblance to the "initial state". Most vehicles will have a `speed_mps` near the speed limit, and the spacing will be regular. This shows that congestion has been avoided. While the final snapshot provides useful information about the system's end state, the primary outputs of our model, such as throughput and average travel time are derived from analyzing the complete sequence of snapshots throughout the simulation.

Bibliography

- [1] Andreas Prinz et al. *Understanding Modelling and Programming*. Springer, 2025. DOI: [10.1007/978-3-031-71280-7](https://doi.org/10.1007/978-3-031-71280-7).